

Table of Contents

File Operations.....	3
Summary.....	3
APT Package Management.....	4
Summary.....	5
C Compilation.....	6
Summary.....	7
Cleanup and Verification.....	8
Summary.....	8

Table of Figures

Figure 1: File Operations using Linux command.....	3
Figure 2: APT Package Management.....	5
Fig 3: C Compilation.....	6
Figure 4: Cleanup and Verification.....	8

File Operations

```
vboxuser@Ubuntu:~/Desktop$ sudo mkdir /linux_lab
[sudo: authenticate] Password:
vboxuser@Ubuntu:~/Desktop$ cd /linux_lab
vboxuser@Ubuntu:/linux_lab$ nano greeting.txt
vboxuser@Ubuntu:/linux_lab$ sudo nano greeting.txt
vboxuser@Ubuntu:/linux_lab$ cat greeting.txt
"Hello Linux!"
vboxuser@Ubuntu:/linux_lab$ cp greeting.txt backup.txt
cp: cannot create regular file 'backup.txt': Permission denied
vboxuser@Ubuntu:/linux_lab$ sudo cp greeting.txt backup.txt
vboxuser@Ubuntu:/linux_lab$ sudo mv backup.txt old_greeting.txt
vboxuser@Ubuntu:/linux_lab$ rm old_greeting.txt
rm: remove write-protected regular file 'old_greeting.txt'?
vboxuser@Ubuntu:/linux_lab$ ls
greeting.txt  old_greeting.txt
vboxuser@Ubuntu:/linux_lab$ rm old_greeting.txt
rm: remove write-protected regular file 'old_greeting.txt'? y
rm: cannot remove 'old_greeting.txt': Permission denied
vboxuser@Ubuntu:/linux_lab$ rm -rf old_greeting.txt
rm: cannot remove 'old_greeting.txt': Permission denied
vboxuser@Ubuntu:/linux_lab$ sudo rm -rf old_greeting.txt
vboxuser@Ubuntu:/linux_lab$ ls
greeting.txt
vboxuser@Ubuntu:/linux_lab$ sudo mkdir temp_dir
vboxuser@Ubuntu:/linux_lab$ sudo rmdir temp_dir
vboxuser@Ubuntu:/linux_lab$ ls -l greeting.txt
-rw-r--r-- 1 root root 15 Sep  8 05:52 greeting.txt
vboxuser@Ubuntu:/linux_lab$ SSS
```

Figure 1: File Operations using Linux command

Summary

At first a new directory was created using the command `mkdir` and navigated to the new directory using `cd` directory name. Then, a new file was created in the nano editor using the command `nano filename.txt` and I wrote something and saved it. After this the file was viewed using the command `cat` and then copy and rename operations were done. At last the file and directories were removed and permission was checked.

APT Package Management

```
vboxuser@ubuntu:/$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:6 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:7 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:8 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.3 kB]
Get:9 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [409 kB]
Get:10 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [79.5 kB]
Get:11 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:12 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:13 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:14 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3,036 B]
Get:15 http://np.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,024 B]
Fetched 2,161 kB in 3s (717 kB/s)
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@Ubuntu:/$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 9
vboxuser@Ubuntu:/$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

vboxuser@Ubuntu:/$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy
```

```
amkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
tools for common BAM file manipulations

eancount/resolute 3.2.0-2build1 all
Double-entry accounting from text files

errynet-dashboard/resolute 3.10.2-2build1 all
deep learning gateway - python3 modules

rltty/resolute,now 6.7-1ubuntu6 amd64 [installed,automatic]
Access software for a blind person using a braille display

budgie-app-launcher-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to provide an alternative means to launch applications

budgie-brightness-controller-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to control the brightness of the screen

budgie-clockworks-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to display clock across multiple time zones

budgie-countdown-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet providing a countdown capability on the Budgie Desktop

budgie-dropby-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to popup when a USB device is connected

budgie-extras-common/resolute-updates 2.2.3-1ubuntu1.1 amd64
Shared component of budgie-extras applets

budgie-extras-daemon/resolute-updates 2.2.3-1ubuntu1.1 amd64
Extras daemon for budgie-extras capabilities
lines 1-44
```

Figure 2: APT Package Management

Summary

In the second part, installed the tools needed for the rest of the lab using the command apt install and apt update and also checked the gcc version.

C Compilation

```
ashim@Ubuntu:/linux_lab$ nano hello.c
ashim@Ubuntu:/linux_lab$
ashim@Ubuntu:/linux_lab$ sudo hello.c
sudo: 'hello.c': command not found
ashim@Ubuntu:/linux_lab$ sudo nano hello.c
ashim@Ubuntu:/linux_lab$ gcc hello.c -o hello_program
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file hello_program: Permission denied
collect2: error: ld returned 1 exit status
ashim@Ubuntu:/linux_lab$ sudo gcc hello.c -o hello_program
ashim@Ubuntu:/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 root root 15992 Sep  8 17:12 hello_program
ashim@Ubuntu:/linux_lab$ sudo gcc -Wall hello.c -o hello_program_warn
ashim@Ubuntu:/linux_lab$ ./hello_program_warn
Hello, _Linux_Lab!
sum_of_10_5_is_15
ashim@Ubuntu:/linux_lab$ sudo nano hello.c
ashim@Ubuntu:/linux_lab$ sudo gcc -Wall hello.c -o hello_program_warn
ashim@Ubuntu:/linux_lab$ ./hello_program_warn
Hello, _Linux_Lab!
sum_of_10_5_is_15
ashim@Ubuntu:/linux_lab$ ./hello_program
Hello, _Linux_Lab!
sum of 100 200 is 300
```

Fig 3: C Compilation

Summary

In this section, writing code, compiling and running a simple c program was done. Moreover the written code was modified and recompiled again.

Cleanup and Verification

```
ashim@Ubuntu:/linux_lab$ cd
ashim@Ubuntu:~$ ls -lh /linux_lab/
total 44K
-rw-r--r-- 1 root root  16 Sep  8 16:20 greeting.txt
-rw-r--r-- 1 root root 148 Sep  8 17:14 hello.c
-rwxr-xr-x 1 root root 16K Sep  8 17:12 hello_program
-rwxr-xr-x 1 root root 16K Sep  8 17:15 hello_program_warn
-rw-r--r-- 1 root root  16 Sep  8 16:21 old_greeting.txt
ashim@Ubuntu:~$ cat /linux_lab/hello.c
#include <stdio.h>

int main() {
    int a=100, b=200;
    printf("Hello, Linux Lab!\n");
    printf("The sum of %d and %d is %d\n",a, b, a+b);
    return 0;
}
ashim@Ubuntu:~$ █
```

Figure 4: Cleanup and Verification

Summary

The cleanup process was done and viewed the content of the c file for the last time.